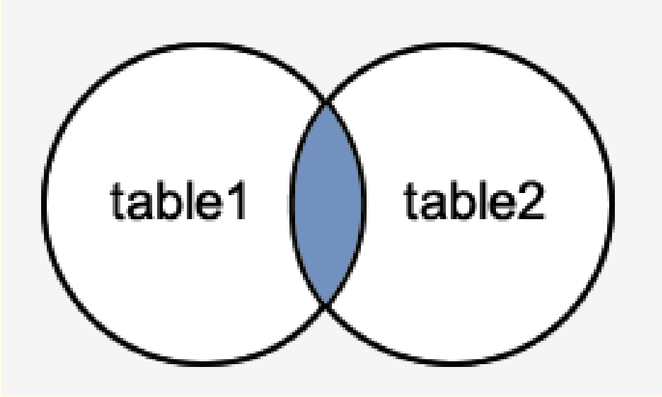
Join clause

Topics List

- JOIN
- Multi-table JOIN
- OUTER JOIN

Join

- A SQL join is an instruction to combine data from two (or more) sets of data (i.e. two tables) based on a related column between them.
- This Join is known as an equijoin or an inner join.




```
SELECT [DISTINCT|ALL]{*|[columnExpression [AS newName]][,...]}
FROM Tablename [alias]
[JOIN Tablename ON ...]
[WHERE condition]
[GROUP BY columnlist][HAVING condition]
[ORDER BY columnList];
```

Join

- The JOIN keyword selects all rows from both tables if there is a match between the join columns in both tables (i.e. Primary Key - Foreign Key link).

```
SELECT column_name(s)
FROM table1
JOIN table2
ON table1.column_name=table2.column_name;
```

Join

customers

customer_id	first_name	last_name	email	address	city	state	zipcode
1	George	Washington	gwashington@usa.gov	3200 Mt Vernon Hwy	Mount Vernon	VA	22121
2	John	Adams	jadams@usa.gov	1250 Hancock St	Quincy	MA	02169
3	Thomas	Jefferson	tjefferson@usa.gov	931 Thomas Jefferson Pkwy	Charlottesville	VA	22902
4	James	Madison	jmadison@usa.gov	11350 Constitution Hwy	Orange	VA	22960
5	James	Monroe	jmonroe@usa.gov	2050 James Monroe Parkway	Charlottesville	VA	22902

Primary Key - Foreign Key link

orders

order_id	order_date	order_amount	customer_id
1	07/04/1776	$234.56	1
2	03/14/1760	$78.50	3
3	05/23/1784	$124.00	2
4	09/03/1790	$65.50	3

Join

```sql
SELECT first_name, last_name, order_date, order_amount
FROM customers
JOIN orders
ON customers.customer_id = orders.customer_id;
```

first_name	last_name	order_date	order_amount
George	Washington	07/4/1776	$234.56
John	Adams	05/23/1784	$124.00
Thomas	Jefferson	03/14/1760	$78.50
Thomas	Jefferson	09/03/1790	$65.50

Join

- The following example selects each member name from the Gymmember table and corresponding (matching) membership from the Membershiptype table:

```
SELECT CONCAT(firstName, " ", lastName) AS Name, membership As Membership
FROM Gymmember JOIN Membershiptype
ON Gymmember.memberType = Membershiptype.membertype
ORDER BY lastname, firstname;
```

Join

- There are 2 variations on the Join statement. Both require that the columns being joined have the same name:
  - USING clause

```
SELECT CONCAT(firstName, " ", lastName) AS Name, membership As Membership
FROM Gymmember JOIN Membershiptype USING (memberType)
ORDER BY lastname, firstname;
```

  - NATURAL JOIN

```
SELECT CONCAT(firstName, " ", lastName) AS Name, membership As Membership
FROM Gymmember NATURAL JOIN Membershiptype
ORDER BY lastname, firstname;
```

Join

- In each SELECT statement that requires a JOIN, there are four things to do:

    1. Identify which tables have the data you are looking for. These table names are used in the JOIN clause.
    2. Identify which columns are the primary and foreign keys in these tables. We use these in the ON clause.
    3. Identify which columns we want the query to output. We put these in the SELECT clause of the statement.
    4. Add any conditions necessary (if there are any).

Join

- When we join 2 tables one record is returned per match, so for example if we join the Gymmember and Membershiptype tables the result can be seen in the output.

```
SELECT  membership As Membership, CONCAT(firstName, " ", lastName) AS Name
FROM Gymmember JOIN Membershiptype
ON Gymmember.memberType = Membershiptype.membertype
ORDER BY membership;
```

Membership	Name
Standard	Sean O'Brien
Standard	Bridget Doyle
Standard	Conor McGrath
Standard	Melissa Reddy
Standard	Mary Fitzgerald
Discount Standard	Aoife Murphy
Discount Standard	Eoin Byrne
Discount Standard	Conor Kelly
Premium	Patrick Kelly
Premium	Ciara O'Connor

Join

- To just return one record per title, we can use GROUP_CONCAT().
- The GROUP_CONCAT() function is used to concatenate data from multiple rows into one field. This is an aggregate (GROUP BY) function that returns a string value if the group contains at least one non-NULL value. Otherwise, it returns NULL.

```
SELECT  membership AS Membership, GROUP_CONCAT(CONCAT(firstName, " ", lastName)) AS Name
FROM Gymmember JOIN Membershiptype
ON Gymmember.memberType = Membershiptype.membertype
GROUP BY membership
ORDER BY membership;
```

Join

Membership	Name
Standard	Sean O'Brien,Bridget Doyle,Conor McGrath,Melissa Reddy,Mary Fitzgerald
Discount Standard	Aoife Murphy,Eoin Byrne,Conor Kelly
Premium	Patrick Kelly,Ciara O'Connor

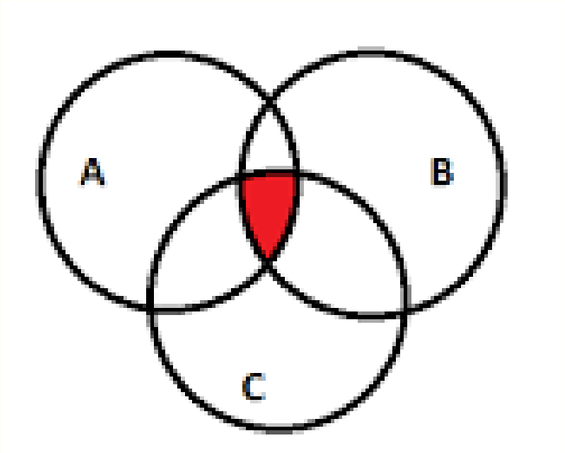
Topics List

- JOIN
- Multi-table JOIN
- OUTER JOIN

Multi-table Join

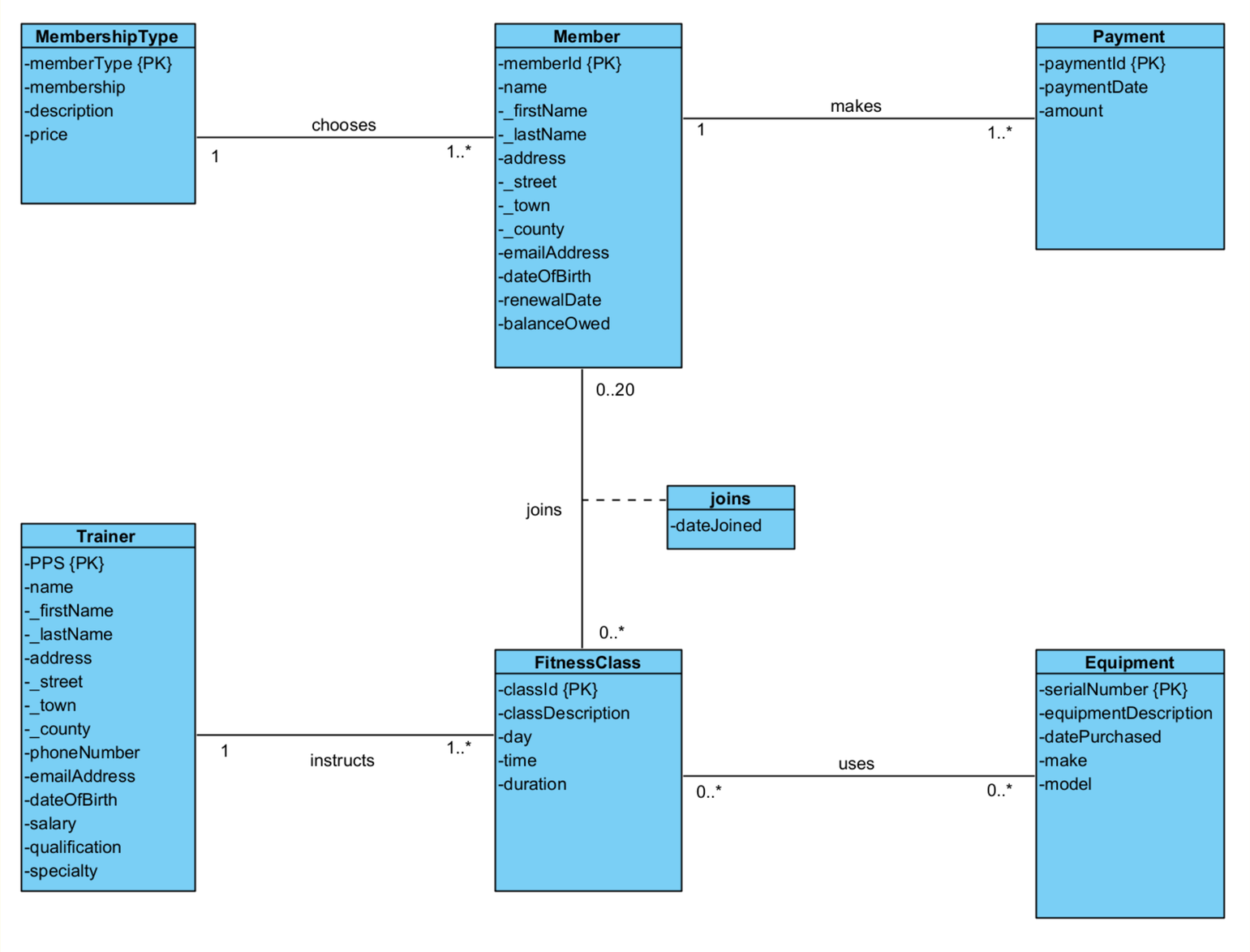
- A Multi-table SQL join is an instruction to combine data from three or more sets of data (i.e. three or more tables).

- Just add another (1 or more) JOIN clause(s) after the first JOIN clause.



Gym example

- Member – FitnessClass
- M:N relationship
- Created memberclass relation at logical mapping stage



Gym example

GymMember

memberId	firstName	lastName
100	Sean	O'Brien
101	Aoife	Murphy
102	Patrick	Kelly
103	Bridget	Doyle
104	Eoin	Byrne
105	Ciara	O'Connor
106	Conor	McGrath
107	Conor	Kelly
108	Melissa	Reddy
109	Mary	Fitzgerald

MemberClass

memberId	classId	joinDate
100	1000	2025-07-02
100	1003	2025-07-02
101	1001	2025-08-10
102	1000	2025-01-21
102	1004	2025-09-01
104	1002	2025-07-09
104	1003	2025-07-02
105	1001	2025-07-02
105	1003	2025-05-02
106	1002	2025-01-03
107	1001	2025-03-02
108	1000	2025-07-02
108	1003	2025-07-02
109	1000	2025-07-02

FitnessClass

classId	classDescription	classDay
1000	Morning Strength Training	Monday
1001	Evening Yoga	Wednesday
1002	Morning Strength Training	Thursday
1003	Yoga at Noon	Friday
1004	HIIT Blast	Friday
1005	Rehab and Stretch	Saturday

Multi-table Join

- For example, for each member we want to return the member's name and the description of each fitness class they are taking.
  1. Identify which tables have the data you are looking for. These table names are used in the JOIN clause. **These are gymmember, fitnessClass, and memberClass.**
  2. Identify which columns are the primary and foreign keys in these tables. We use these in the ON clause. **gymmember – memberClass (memberId); fitnessClass – memberClass (classId).**
  3. Identify which columns we want the query to output. We put these in the SELECT clause of the statement. **firstname, lastname, classDescription**
  4. Add any conditions necessary (if there are any).

Multi-table Join

```sql
SELECT CONCAT(firstname, " ", lastName) AS Member, classDescription Class
FROM gymmember JOIN memberClass
ON gymmember.memberId = memberClass.memberId
JOIN fitnessClass
ON fitnessClass.classId = memberClass.classId
ORDER BY lastname, firstname;
```

Topics List

- JOIN
- Multi-table JOIN
- OUTER JOIN

Outer Join

- Outer joins. When performing an inner join, rows/records from either table that are unmatched in the other table are not returned.

- In an outer join, unmatched rows in one or both tables can be returned.

Outer Join

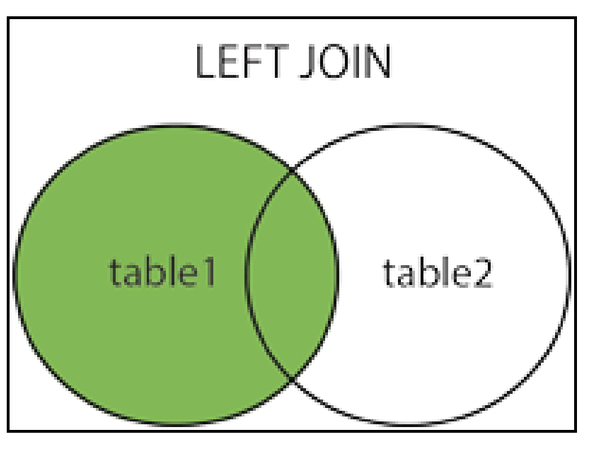
- To perform an outer join in MySQL, we use either the LEFT JOIN or the RIGHT JOIN.

- LEFT JOIN returns only unmatched rows from the left table.

- RIGHT JOIN returns only unmatched rows from the right table

Left Join

- The LEFT JOIN keyword returns all rows from the left (first) table (table1), with the matching rows in the right table (table2).
- The result is NULL in the right side when there is no match.



Left Join

customers

customer_id	first_name	last_name	email	address	city	state	zipcode
1	George	Washington	gwashington@usa.gov	3200 Mt Vernon Hwy	Mount Vernon	VA	22121
2	John	Adams	jadams@usa.gov	1250 Hancock St	Quincy	MA	02169
3	Thomas	Jefferson	tjefferson@usa.gov	931 Thomas Jefferson Pkwy	Charlottesville	VA	22902
4	James	Madison	jmadison@usa.gov	11350 Constitution Hwy	Orange	VA	22960
5	James	Monroe	jmonroe@usa.gov	2050 James Monroe Parkway	Charlottesville	VA	22902

orders

order_id	order_date	order_amount	customer_id
1	07/04/1776	$234.56	1
2	03/14/1760	$78.50	3
3	05/23/1784	$124.00	2
4	09/03/1790	$65.50	3

Left Join

```sql
SELECT first_name, last_name, order_date, order_amount
FROM customers
LEFT JOIN orders
ON customers.customer_id = orders.customer_id;
```

first_name	last_name	order_date	order_amount
George	Washington	07/04/1776	$234.56
John	Adams	05/23/1784	$124.00
Thomas	Jefferson	03/14/1760	$78.50
Thomas	Jefferson	09/03/1790	$65.50
James	Madison	NULL	NULL
James	Monroe	NULL	NULL

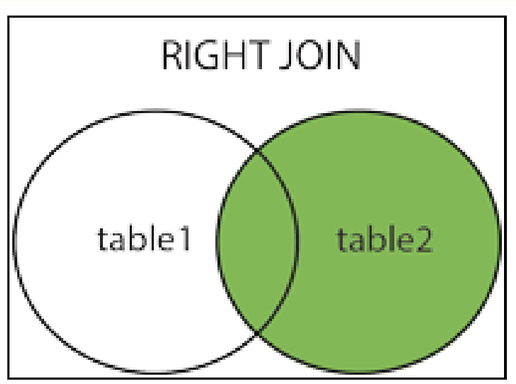
Left Join

- In the following example, we select the member's name and the classDescription of the fitness classes they are taking. If the member is taking no classes, their name is returned and the classDescription value will be NULL.

```sql
SELECT CONCAT(firstname, " ", lastName) AS Member, classDescription Class
FROM gymmember LEFT JOIN memberClass
ON gymmember.memberId = memberClass.memberId
LEFT JOIN fitnessClass
ON fitnessClass.classId = memberClass.classId
ORDER BY lastname, firstname;
```

Right Join

- The RIGHT JOIN keyword returns all rows from the right (second) table (table2), with the matching rows in the left table (table1).
- The result is NULL in the left side when there is no match.



Right Join

- Here is the previous example written with the RIGHT OUTER JOIN syntax. This time all the records from the second (right) table are returned.

```
SELECT CONCAT(firstname, " ", lastName) AS Member, classDescription Class
FROM fitnessClass JOIN memberClass
ON fitnessClass.classId = memberClass.classId
RIGHT JOIN gymmember
ON gymmember.memberId = memberClass.memberId
ORDER BY lastname, firstname;
```